

Jinwoo Lee

Full Stack Engineer

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PROFILE

Full stack engineer with 5+ years of experience shipping production web apps across advertising, Web3, travel tech, insurance, and mental health. Strong in React/Next.js, TypeScript, and creative front-end work (Three.js, WebGL, Mapbox/deck.gl), backed by solid Go, Node, and AWS experience. Currently at DDB NZ, where I lead front-end on Fantasy Herd and deliver high-profile interactive campaigns for McDonald's NZ, Škoda NZ, and Volkswagen NZ.

EXPERIENCE

Front End Engineer • McCann NZ Jan. 2025 – Present

McCann is a global creative agency; the NZ studio builds digital campaigns and interactive experiences for brands including McDonald's NZ, Škoda NZ, and Volkswagen NZ.

- **Škoda NZ — Codes, Word of the Day:** Built a 90-day daily word-puzzle promotional game offering a trip for two to the Tour de France as the grand prize.
- Built the full client app in Next.js 16 / React 19 with Zustand for state, Motion (Framer Motion) for cinematic intro, letter-slot, and shake animations, and Tailwind 4 + shadcn/ui for the design system.
- Implemented a custom letter-slot input system (static vs editable slots, auto-advance, Tab cycling, color-coded states) with responsive sizing for 5–9+ character words across mobile, tablet, and desktop.
- Built the type-safe API layer with Orval + Zod codegen from Swagger, handled 200/204/409/410/500 status flows including graceful 409 "already entered" handling, and used localStorage scoped by campaign date with auto-cleanup for the 3-attempts-per-day mechanic.
- **Volkswagen NZ — VW-Buzz interactive experience:** Developed an interactive map + 3D promotional experience for Volkswagen NZ in React with Mapbox, deck.gl, and Three.js.
- Built an advanced Mapbox UI with react-map-gl and deck.gl overlays (animated rings, labels, paths) supporting multiple map modes (bees / range / floral) with camera transitions, animated fly-to, and Turf-powered geospatial calculations.
- Implemented a custom metaball / "gooey" 3D effect with React Three Fiber with custom shaders, including physics (gravity, drag) and pointer interaction for animated blobs.
- Architected a clean component structure (atoms / molecules / screens / store) with Zustand state management and Radix + Framer Motion for UI.
- **Fantasy Herd — Meadow Fresh's first fantasy-football themed game with 20k NZD prize:** Co-Authored a 5-month Next.js mainly on caching and performance improvements aspect of the web app (Nov 2025 – Mar 2026); #1 individual contributor with 416 commits (24.8% of the project) and 88 Jira tickets delivered across frontend, backend, and DevOps work.
- Owned the core UI architecture: responsive mobile-first layouts, typography system, shared zustand stores/hooks, and a prototype component library (colors, buttons, forms, typography) seamlessly used across the app.
- Built major feature areas end-to-end, including invitational league, global/local league, cached cow data fetching/selection (pre-season / in-season / post-season), Mobile Scan/Barcode, a real-time ticker banner and notification persistence system.
- Led state and session management work: store cleanup, 20-minute inactivity timeout, and a user suspension flow driven by site settings, with backend integration for league join/update logic, scan/barcode wiring, and suspension endpoints.
- Drove performance and quality improvements: improved product lookup speed up $O(1)$ for the mobile scan feature, lazy-loaded MP3 assets, added image caching, migrated the app to Next.js 16 with cache-TTL tuning, and resolved cross-browser issues on Firefox and iOS WebKit.
- **McDonald's NZ — Web app platform:** Built and maintain a Next.js 16 / React 19 web app embedded via webview in the McDonald's NZ mobile app, powering multiple promotional mini-experiences used by customers nationwide.
- Shipped 7+ feature flows including **Perfect Pour** - L&P promotional game with SVG with framer motion animations, **Merch** (loyalty-points redemption for physical merch and digital vouchers), **Coke** and **FIFA** sweepstakes/instant-win campaigns, a **Friends**-themed personality quiz, a **Supporter of the Day** voucher flow, and a general-purpose **voucher scanner/redeemer** with smart routing.

- Integrated with the McDonald's native bridge (Mcdbridge) for auth, loyalty points, QR scanning, fullscreen mode, and deep linking between web and native layers.
- Established the app's engineering foundations: auto-generated API client with Orval + Swagger, CAS Middleware authentication, a shared UI library (Carousel, Drawer, Splash, TopBar, Confetti), and end-to-end test coverage with Playwright.

Front End Engineer • Mobility Labs Aug. 2024 – Jan. 2025

- Developed an in-house geospatial visualisation tool, vista.gl, that processes GEOJSON datasets to create configurable, shareable maps for transport modellers and planners across New Zealand.

Full Stack Engineer • Ion Protocol Mar. 2023 – Mar. 2024

Ion Protocol is an open-source protocol supporting Ethereum stakers and restakers.

- Developed Ion's web app for LST lenders and borrowers using Next.js, TailwindCSS, Node.js, and Postgres, with responsive modular layouts and light/dark mode support.
- Implemented Viem for contract and wallet interactions, and configured the app to support multiple LST assets (weETH, rsETH, and others).
- Deployed Tenderly/Anvil instances for local development and testing environments.
- Implemented application-wide alerting, secure Metamask wallet interactions, and custom hooks for Ion's Solidity contracts plus UniswapV3, BalancerV2 (leverage strategies), and Chainlink price oracles (USD/ETH conversion).
- Built hourly accounting oracles in Go, deployed to AWS Lambda, to monitor EL/CL Ethereum reserves for multiple LST providers (Swell, Lido, fxETH) for Proof-of-Reserve and to monitor validators for slashing and penalties.

Front End Engineer • Serko Jan. 2023 – July 2023

Serko is Booking.com's subsidiary managing corporate travellers on the Booking.com platform.

- Developed a design-token framework integrating Figma into Storybook, streamlining the workflow between the design team and engineers.
- Upgraded Storybook to leverage MDX, unlocking new features to improve documentation, component usability, and cross-team collaboration.

Technology Consultant • Datacom Nov. 2021 – Dec. 2022

- Collaborated with IAG (NZ subsidiary of Insurance Australia Group) to develop SpringBoot APIs automating insurance pricing in New Zealand.
- Proposed and implemented a proof-of-concept 3D interactive visualisation tool for Heliase, a start-up using a 3D force-directed graph to visualise relationships between patent documents for researchers.
- Optimised the patent visualisation tool using client/server-side caching and a particle system with Next.js, Three.js, and GraphQL to reduce browser load.

Software Engineer • Clearhead Nov. 2020 – Nov. 2021

Clearhead is a start-up providing self-help tools for mental health, wellbeing insights for companies, and therapist matching for users.

- Developed the initial front-end unit test suite to improve code reliability.
- Built a blue-green deployment pipeline with Terraform to improve DevOps robustness and uptime.
- Created a comprehensive Storybook UI component library for the Chatbot for seamless integration into the app.

SKILLS

Frontend: React, Next.js, TypeScript, TailwindCSS, shadcn/ui, Zustand, Motion/Framer Motion, Storybook

Creative / Visualisation: Three.js, React Three Fiber, WebGL/GLSL, D3.js, Mapbox, deck.gl, Turf

Backend: Node.js, Go, SpringBoot, GraphQL, PostgreSQL, MongoDB

Web3: Viem, Solidity contract integration, Metamask, Tenderly/Anvil, Chainlink oracles

Tooling & Infra: AWS (Lambda, Solutions Architect Associate), Terraform, Playwright, Biome, Orval + Zod, Supabase

EDUCATION

Bachelor of Computer Science — Auckland University of Technology 2018 – 2021

CERTIFICATES

- **AWS Solutions Architect Associate** — AWS
- **IBM Data Science Professional Certificate** — Coursera

